

Choosing The Right EMBEDDED DEVELOPMENT BOARD



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Development and prototype boards are computer boards that are used to develop or test electronic modules. These are used to evaluate programs for embedded devices such as controllers, point-of-sale (PoS) terminals, kiosks and information appliances.

Development boards combine a processor, chipset, memory and onboard peripherals with debugging features. Specifications for development boards include bus type, processor type, form factor, number of ports, port type, memory and operating system.

Factors to keep in mind

There are a number of conventional benchmarking factors that we all use in sorting through alternatives. These include capabilities of the target controller (speed,

power consumption and on-chip peripherals), available features (memory being the major consideration along with supported communication protocols), onboard peripherals and technical support (compiler tool chain and sample applications) from the vendor.

There are a number of other considerations that should drive us while looking at development boards. Described below are a few of these:

Open design. Take any popular development board that has been registering an active user community, timely enhancements and hundreds of developers working towards porting new applications to the board. Openness in a design has got more to do with the mental satisfaction of the buyer.

What is the general selection criteria for MCU based development boards

Serviceability. It is important that the boards can be repaired if and when something goes wrong. To this end, packaging of the MCU needs to be considered. It is important to consider whether the MCU is a through-hole part-mounted in a socket or is it surface-mounted. And if it is surface-mounted, can it be replaced when needed?

Pedagogy. Educational value and cognitive learning effects of the board must be taken into account. Big and fancy development boards may offer a tremendous amount of features and look really cool, but if these features diminish what the student understands of the core technology, then these may be detrimental to the cause.

Reliability. One of the main goals of development boards is to ensure that we have an operational development board for each and every experiment. Any time spent attempting to fix the board can take away a lot of critical and important time.

Programming support/options. Newer MCUs allow in-circuit programming while these are located in the system. Options may include an external programmer or a self-programming mechanism.

Processor support (single or multiple). If one is working on multiple controller families or different part numbers, this factor becomes critical because having separate development boards for each controller may not be feasible all the time. A proper decision to this effect can save money.

Cost. Obvious factor; compared to the value to the user, costs should be at par.

Design control. There is nothing more frustrating than selecting an appropriate development board only to have the manufacturer



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make a change that ultimately forces a new development board to be selected. It is therefore necessary to have some type of design control over the development board that is ultimately chosen.

Debugging capabilities. Depending upon the kind of work to be done on the development board, JTAG, single-wire debug or whatever else is necessary to enable users of easy debugging and faster development should be considered.

System clock options. Whether the board gives multiple clock options such as internal RC oscillator plus external crystal installed or only one can make or break some of the developments. This is not very critical as compared to all other factors.

Form factor. While development boards are ultimately used for development purpose, their form factor is critical. A board that is much bigger in size and difficult to move around may not be preferred, while a small-size handy board that can be taken anywhere in the pocket or a small box could help engineers develop things whenever they want.

Readiness for final integration. Like I said before, many a time, development boards are just a perfect fit for the final product and can have everything that the final product requires. In such cases, making another board may not be warranted (except where outline of board, form-factor, etc are different). Thus, the development board can directly go into the product box. In such cases, this becomes a very important factor since it has high potential to save manufacturing money and time.